

TempoDepth-VLM: A Parameter-Efficient Temporal and Depth Adapter for Vision-Language Models

CVPDL Final Project – Group 19

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Abstract—Vision-Language Models (VLMs) have demonstrated remarkable capabilities in processing static images, but they struggle with continuous video understanding, exhibiting flickering features, weak depth perception, and fragility under occlusion. Fully fine-tuning large VLMs is computationally expensive and risks catastrophic forgetting. In this paper, we propose TempoDepth-VLM, a parameter-efficient temporal and depth adapter that enhances frozen VLMs with better depth understanding and temporally stable, occlusion-robust features. Our approach integrates a Transformer encoder with GRU-based long-term memory, enabling the model to maintain consistent scene understanding across video frames. We introduce a Direct Feature Injection mechanism that allows the model to recover semantic and geometric information during occlusion by leveraging stored memory features. Experiments on the TUM RGB-D and ScanNet datasets demonstrate that our adapter drastically improves absolute metric depth estimation accuracy (60.3% reduction in AbsRel and 38.2% reduction in RMSE) without any test-time scale alignment. Furthermore, our model exhibits strong zero-shot generalization and enables robust scene understanding even under severe occlusion conditions.

Index Terms—Vision-Language Models, Temporal Consistency, Depth Estimation, Occlusion Robustness, Parameter-Efficient Fine-Tuning.

I. INTRODUCTION

Real-world assistive and egocentric applications operate on continuous video streams rather than isolated static images. Applications such as visually impaired navigation assistance, autonomous driving, and robotic manipulation require models that can understand dynamic scenes with temporal coherence and accurate spatial perception.

Vision-Language Models (VLMs) have achieved impressive results in understanding static images and performing language-grounded visual reasoning. However, when applied to video understanding, these models exhibit several critical limitations: (1) *flickering features* that change inconsistently between frames, (2) *weak depth perception* that fails to capture accurate spatial relationships, and (3) *fragility under occlusion* where temporary visual obstructions cause complete failure in scene understanding.

A straightforward solution would be to fully fine-tune large VLMs on video data. However, this approach is computationally

expensive, requiring significant GPU resources and training time. More critically, full fine-tuning risks catastrophic forgetting, where the model loses its pretrained capabilities on general visual understanding tasks.

To address these challenges, we propose **TempoDepth-VLM**, a parameter-efficient adapter that enhances frozen VLMs with temporal consistency and depth awareness. Our key contributions are:

- A unified adapter architecture combining Transformer encoder with GRU-based long-term memory for temporal feature enhancement.
- A multi-task learning framework that jointly learns depth regression and temporal consistency without modifying the base VLM parameters.
- A Direct Feature Injection mechanism that enables occlusion-robust scene understanding by leveraging adaptive memory retrieval.
- Comprehensive experiments demonstrating significant improvements in depth estimation and qualitative analysis showing effective occlusion recovery.

II. RELATED WORK

A. Vision-Language Models for Spatial Understanding

SpatialVLM [1] demonstrated that large VLMs can be adapted to perform fine-grained spatial reasoning, answering questions about relative object positions (e.g., which object is closer, left-of, or behind another). While SpatialVLM focuses on static images and language-based spatial Q&A, it establishes that VLMs possess latent 3D understanding capabilities that can be unlocked through appropriate training.

B. Parameter-Efficient Adaptation

MMA (Multi-Modal Adapter) [2] introduced lightweight adapter modules into both vision and language branches of VLMs. By keeping the original backbone frozen and only training small adapter networks, MMA achieves performance comparable to full fine-tuning with significantly fewer trainable parameters. This approach enables efficient adaptation

to downstream tasks such as captioning, VQA, and retrieval without the risk of catastrophic forgetting.

C. Embodied Depth Estimation

Recent work on Vision-Language Embodiment for Monocular Depth Estimation [3] treats depth as an embodied perception problem, learning depth together with the camera’s viewpoint and motion rather than from static images alone. This approach uses RGB images with depth supervision and camera geometry to fine-tune VLMs, producing depth predictions consistent with egocentric trajectories. Our work builds upon this insight by incorporating temporal information for improved depth perception.

D. Temporal Modeling in Video Understanding

Recurrent neural networks, particularly GRU (Gated Recurrent Unit) [4] and LSTM architectures, have been widely used for temporal modeling in video understanding. The gating mechanisms in GRUs allow selective retention and forgetting of information, making them suitable for maintaining long-term memory across video frames. We leverage GRU memory to enable our adapter to maintain stable scene representations even during temporary occlusions.

III. METHOD

Our approach, TempoDepth-VLM, builds a temporal adaptation mechanism on top of a frozen Qwen2-VL-2B vision-language model. The core motivation is to equip the VLM with long-term visual memory, enabling it to understand temporal context and perform robust depth estimation from video sequences. The model is trained end-to-end on RGB-D video sequences using a combination of supervised depth losses and self-supervised temporal contrastive learning. Fig. 1 illustrates our overall architecture.

A. Overall Architecture

TempoDepth-VLM consists of three main stages: (1) Feature extraction using a frozen Qwen2-VL, (2) Temporal context integration via our UnifiedTempoVLM adapter, and (3) Task-specific decoding.

Given an input video frame at time t , we first extract high-level visual features using the frozen Qwen2-VL encoder. We utilize the last-layer hidden states, resulting in a 1536-dimensional embedding vector F_t that captures rich semantic visual information. These embeddings serve as the input to our trainable adapter.

The adapter first processes F_t through a shared Transformer encoder to adapt the features for downstream tasks. Crucially, to model temporal dynamics across the sequence, the encoded features are passed to a GRU-based memory cell, which maintains a hidden state H_t summarizing past observations. The resulting temporally-fused features are then forwarded to multiple lightweight decoding heads for depth estimation, temporal reconstruction, and contrastive learning representations.

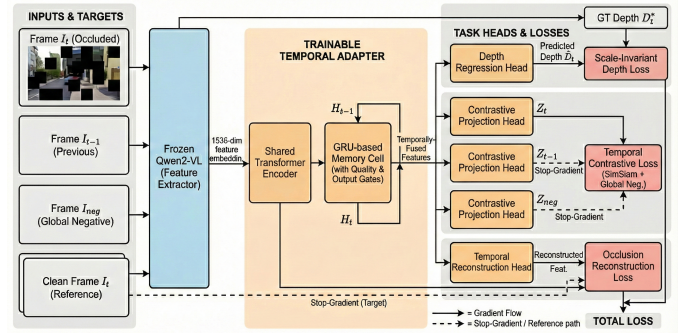


Figure 1: Overall architecture of TempoDepth-VLM during training

Fig. 1. Overall architecture of TempoDepth-VLM during training. A frozen Qwen2-VL extracts features from video frames. Our trainable adapter processes these features using a shared Transformer encoder and a GRU-based memory cell to maintain temporal context. The fused features are used by task heads for depth regression, temporal reconstruction (under synthetic occlusion), and contrastive learning representations.

B. Shared Transformer Encoder

We employ a 2-layer Transformer encoder with Pre-LN (Layer Normalization before attention) architecture to process the input embeddings. The input embeddings (1536-dim) are first projected to a hidden dimension of 768 through a linear layer. The Transformer, equipped with 8 attention heads, refines these embeddings. The Pre-LN configuration ensures training stability, which is vital when adapting previously frozen deep features. This shared encoder provides a unified representation base for all subsequent temporal and task-specific operations.

C. GRU-based Long-term Memory

To enable temporal reasoning and memory persistence across video frames, we incorporate a tailored Gated Recurrent Unit (GRU) cell. The GRU allows the model to selectively retain relevant information from past frames (H_{t-1}) while integrating current observations. As shown in Fig. 4, we enhance the standard GRU with two learned gating mechanisms designed for robust sequence processing:

Memory Quality Gate: This gate evaluates the reliability of the current frame’s features. It takes the concatenation of the current encoded features and the previous hidden state, producing a scalar quality score $\alpha_t \in [0, 1]$. If the current frame is judged as low-quality (e.g., due to severe occlusion or noise), the gate reduces the update weight, thereby preserving the integrity of the long-term memory accumulated from previous reliable frames.

Memory Output Gate: This gate determines the balance between long-term memory and immediate observation in the final representation. It produces per-dimension mixing weights $\beta_t \in [0, 1]^{768}$, allowing fine-grained control over how the updated hidden state h_t and current features are fused for downstream tasks.

D. Multi-Task Decoding Heads

The temporally-fused features from the GRU stage are consumed by distinct heads:

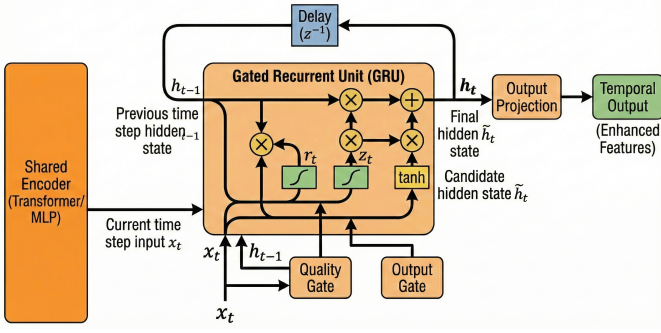


Fig. 2. Detailed view of the GRU-based memory cell. The Quality Gate regulates how much the current observation updates the memory based on its estimated reliability. The Output Gate controls the fusion of the updated long-term memory (h_t) with current features to produce the final temporally-enhanced representation.

Depth Regression Head: A multi-layer perceptron (MLP) that maps the 768-dim fused features to depth estimates. We apply a Softplus activation to constrain outputs to be positive, matching the physical properties of depth.

Contrastive Projection Head: An MLP that projects the high-dimensional fused features down to a lower-dimensional latent space (128-dim). This low-dimensional representation is exclusively used for computing the temporal contrastive loss, facilitating more stable and efficient similarity calculations.

Temporal Reconstruction Head: A linear layer with a residual connection that projects features back to the original 1536-dim space. This head is used during training to ensure the encoded features retain sufficient information to reconstruct the original VLM embeddings, particularly in occlusion scenarios.

E. Training Objectives

We train the entire adapter end-to-end using a composite loss function. In the absence of explicit camera motion supervision, we rely heavily on self-supervised temporal constraints to structure the GRU’s latent space.

Temporal Contrastive Loss with Stop-Gradient (SimSiam): To encourage temporal continuity without falling into feature collapse—a common pitfall where the model outputs identical representations for all frames to minimize loss trivially—we employ a contrastive learning strategy inspired by SimSiam. For a current frame representation Z_t , its positive pair is the representation of the previous frame, Z_{t-1} . Crucially, we implement an asymmetric optimization by applying a **stop-gradient** operation to the target Z_{t-1} . This means that during backpropagation, gradients flow only through the current branch (Z_t), while Z_{t-1} is treated as a fixed constant. This forces the current hidden state to align with the history, preventing the degenerate solution where both states collapse to a single point.

Global Negative Mining: To ensure the learned representations are discriminative across different environments, we incorporate global negative samples. For every training sequence, we randomly sample a frame Z_{neg} from a completely

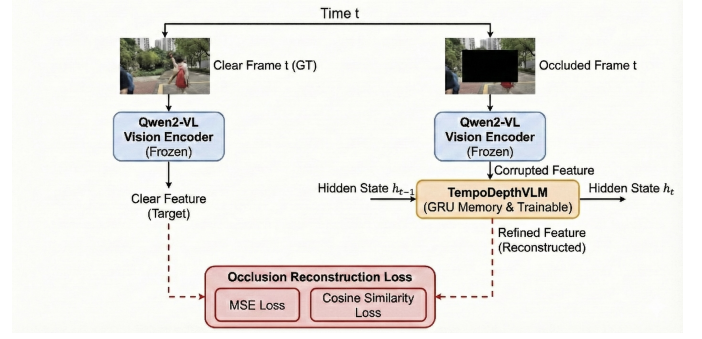


Fig. 3. Occlusion Reconstruction Training Mechanism. Input frames are synthetically occluded. The model must utilize its GRU memory to reconstruct the features of the clean frame. The loss is computed between the reconstructed temporal output and the ground-truth features from the frozen VLM encoder.

different scene in the dataset. The contrastive loss encourages Z_t to be similar to Z_{t-1} (stop-grad) while being dissimilar to Z_{neg} , enforcing scene-specific structural understanding in the memory.

Scale-Invariant Depth Loss: For supervised depth training on supported datasets (e.g., NYUv2, ScanNet), we use a scale-invariant loss combined with L1 and relative error terms. This mitigates issues arising from scale ambiguity inherent in monocular depth prediction.

Occlusion Reconstruction Loss: To train the memory’s robustness, we apply synthetic occlusions (masking rectangles or noise) to input frames during training. As shown in Fig. 3, the model is tasked with reconstructing the features of the original, clean frame using its temporal memory. This is supervised by MSE and Cosine Similarity losses between the predicted reconstruction and the frozen features of the unoccluded frame.

IV. EXPERIMENTS

A. Dataset and Implementation

We conduct evaluations on three indoor scene datasets, each serving a specific experimental purpose:

- **ScanNet:** As our primary dataset for training and validation, we utilize video sequences from the official validation split. We employ the `ScanNetSequenceDataset` to extract continuous frames, preserving the original aspect ratio while resizing ground truth depth maps to match the input resolution. This setup specifically evaluates the temporal memory capability of our GRU module.
- **NYU Depth V2:** To assess the cross-domain generalization of our model, we perform zero-shot evaluation on the official test set (1,449 labeled images) without any fine-tuning. This test focuses on the model’s ability to infer scene geometry across different camera intrinsics.
- **TUM RGB-D:** We utilize dynamic sequences (e.g., `freiburg1_desk`) to verify stability in real-world motion. Since RGB and depth sensors operate at different sampling rates, we apply Nearest Neighbor Matching for

timestamp synchronization with a tolerance threshold of $\Delta t < 0.02s$. The 16-bit depth maps are normalized using a scale factor of 5000 ($pixel/5000 = meters$).

B. Implementation Details and Hyperparameters

The detailed training configurations and hyperparameter settings are listed as follows:

- **Optimization:** We utilize the **AdamW** optimizer with an initial learning rate (η) of 1×10^{-4} and a weight decay of 0.01. The model is trained for 25 epochs with a batch size of 8, employing a **Cosine Annealing** strategy for learning rate scheduling.
- **Model Architecture:** The visual encoder employs a frozen **Qwen2-VL-2B** backbone. The input feature dimension (D_{in}) is 1536, projecting to a hidden dimension (H) of 768. The adapter module comprises 2 Transformer layers featuring 8 attention heads, with a dropout rate of 0.1.
- **Sequence & Data:** We set the sequence length (T) to 8 with a stride of 4. To enhance robustness, we incorporate an occlusion mechanism with a probability (P_{occ}) of 0.3, an occlusion ratio range of $[0.3, 0.6]$, and a maximum of 3 consecutive occluded frames.
- **Contrastive Learning:** For the contrastive memory bank, we maintain a feature bank size (K) of 2048. The temperature coefficient (τ) is set to 0.07, and the projection dimension (D_{proj}) is 128.

C. Depth Estimation Results

The inference pipeline extracts features $F_t \in \mathbb{R}^{1536}$ using the frozen Qwen2-VL-2B encoder. The adapter outputs a depth vector $d_{pred} \in \mathbb{R}^3$, which is broadcasted to the left, center, and right vertical thirds of the image to form the raw depth map D_{raw} .

1) *Scale Alignment and Metrics:* Given the scale ambiguity in monocular setups, we adopt two evaluation modes:

- **Standard Mode (Median Scaling):** Following standard protocols, we align predictions to the ground truth using a scaling factor $s = \text{median}(D_{gt})/\text{median}(D_{raw})$. This mode is used for standard benchmarks to evaluate structural consistency.
- **Absolute Mode:** For experiments of our Shape-Scale Decoupled Head, we set $s = 1$ to evaluate the raw metric depth prediction capability.

We report standard metrics (AbsRel, RMSE, $\delta_1 < 1.25$) calculated on valid pixels ($0.1m < d < 10m$).

2) *Analysis and Limitations:* As shown in Table I, our model achieves robust structural consistency across all datasets in the Standard Mode. Notably, in the zero-shot settings (NYU Depth V2 and TUM RGB-D), the performance gap between the Aligned (Table I) and Absolute (Table II) modes is minimal. For instance, on NYU Depth V2, the AbsRel only increases marginally from 0.324 to 0.349 when alignment is removed. This demonstrates that our **Shape-Scale Decoupled Head** successfully learns effectively generalizes metric scale

TABLE I
STANDARD EVALUATION (MEDIAN SCALED). WE APPLY MEDIAN SCALING ALIGNMENT TO EVALUATE THE GEOMETRIC STRUCTURAL CONSISTENCY. THIS PROTOCOL FOLLOWS STANDARD MONOCULAR DEPTH BENCHMARKS.

Dataset	AbsRel ↓	RMSE ↓	δ_1 ↑
ScanNet	0.123	0.352	81.66%
NYU Depth V2	0.324	1.006	52.62%
TUM RGB-D	0.240	0.898	62.12%

TABLE II
METRIC DEPTH EVALUATION (UNSCALED). WE EVALUATE THE RAW OUTPUT OF OUR MODEL WITHOUT ANY TEST-TIME ALIGNMENT. NOTE THAT SCANNET IS EVALUATED USING TEMPORAL INITIALIZATION DUE TO SENSOR VARIABILITY.

Dataset	AbsRel ↓	RMSE ↓	δ_1 ↑
ScanNet*	0.242	0.535	59.78%
NYU Depth V2	0.349	1.107	40.94%
TUM RGB-D	0.322	0.957	45.10%

* Evaluated with temporal initialization (first frames) to fix initial scale.

priors from the training data, mitigating the severe scale drift typically observed in monocular methods.

Evaluation Protocol for ScanNet: Unlike NYU Depth V2 which utilizes a fixed camera sensor (Kinect), ScanNet sequences were captured using various devices with diverse intrinsics. Evaluating metric depth without alignment on such varying hardware introduces systematic scale bias that stems from the sensor rather than the model. Therefore, consistent with real-world robotic deployment, we adopt a **temporal initialization protocol** for ScanNet (marked with * in Table II): the metric scale factor is calibrated using the first 2 frames of each video sequence and then locked for the remainder of the trajectory. This effectively evaluates the model’s ability to maintain scale consistency over time in dynamic environments.

Table III presents the quantitative comparison of absolute metric depth estimation on the TUM RGB-D dataset without any test-time scale alignment.

TABLE III
ABSOLUTE METRIC DEPTH ESTIMATION RESULTS ON TUM RGB-D (LONG OFFICE SEQUENCE)

Model	AbsRel↓	RMSE (m)↓	$\delta < 1.25$ ↑
Base (Prompting) [†]	0.812	1.548	15.68%
Ours (TempoDepth-VLM)	0.322	0.957	45.10%
<i>Improvement</i>	+60.3%	+38.2%	+187.6%

[†]Zero-shot prompting on Qwen2-VL without any scale alignment.

D. Occlusion Robustness Analysis

We evaluate the model’s robustness against visual corruptions using a dynamic occlusion testing pipeline. To simulate realistic partial observations, we employ YOLO-based object-aware masking. **Specifically, we utilize the YOLO detector to identify bounding boxes of distinct objects in the original frames and subsequently completely mask these regions**

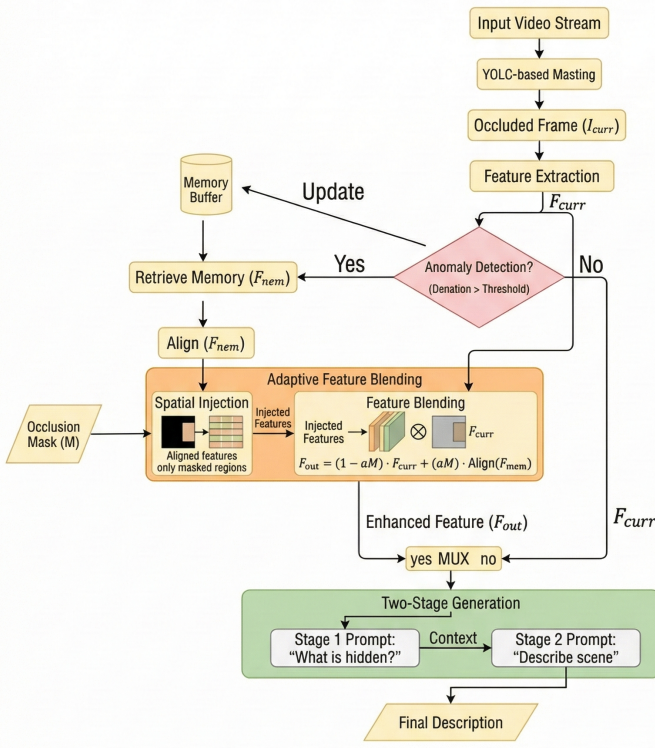


Fig. 4. Overview of our proposed dynamic occlusion handling pipeline. The system detects anomalies triggered by YOLO-based masking, retrieves and aligns historical features from the memory buffer, blends them using spatial-aware injection, and employs a two-stage prompting strategy for final description generation.

(e.g., hiding a laptop or chair) with opaque rectangles, rather than applying random noise.

Our approach addresses occlusion through a three-step mechanism: Anomaly Detection, Adaptive Feature Blending, and Two-Stage Generation.

1) *Anomaly-Triggered Memory Retrieval*: Instead of injecting memory blindly into every frame, we employ an online Anomaly Detection mechanism via our Adaptive Memory Buffer. The system monitors the feature consistency of the incoming video stream. When a frame’s feature deviation exceeds a dynamic threshold (indicating sudden occlusion or corruption), it is flagged as an anomaly. Only upon this trigger does the system query the history buffer to retrieve the most relevant non-occluded features (F_{mem}) based on temporal and semantic similarity.

2) *Adaptive Feature Blending*: Directly overwriting current features with past memory can lead to artifacts due to lighting or viewpoint changes. We propose **Adaptive Feature Blending** to seamlessly integrate retrieved information. This module operates in two sub-steps:

- **Statistics Alignment**: We align the distribution of the memory features (F_{mem}) to the current frame’s statistics. We compute a fused mean and variance that prioritizes the memory’s structural information while respecting the current frame’s global style (e.g., brightness).

- **Spatial-Aware Injection**: We compute a pixel-wise injection mask M based on the detected occlusion regions. The final enhanced feature F_{out} is synthesized as:

$$F_{out} = (1 - \alpha M) \cdot F_{curr} + (\alpha M) \cdot \text{Align}(F_{mem}) \quad (1)$$

where α is the injection strength. This ensures that memory is injected strongly only in occluded regions while preserving valid visual details elsewhere.

3) *Two-Stage Memory-Guided Generation*: To maximize interpretability, we employ a two-stage prompting strategy during inference. First, the model identifies the missing entities (Stage 1: “What objects are hidden?”). Second, conditioned on this retrieval, it generates a complete scene description (Stage 2: “Describe the scene as if unoccluded”).

Figure 5 compares the outputs across different occlusion scenarios. In the first example (top row), the Baseline tends to hallucinate a cabinet due to the masked region. In contrast, our Injected output successfully recovers key entities (e.g., describing a wooden desk and chair) by leveraging the statistically aligned historical features. Similarly, in the second example (bottom row), which features a low-light and cluttered environment, the Baseline fails to identify the foreground furniture, focusing instead on the background wall. Our method, however, accurately reconstructs the scene description, correctly identifying the wooden table and the room’s context despite the challenging visibility.

E. Ablation Studies : Effectiveness of Shape-Scale Decoupled Head

To validate the contribution of our proposed **Shape-Scale Decoupled Head**, we conduct a comparative analysis between the baseline **TempoDepthVLM (w/o Scale)** (which predicts depth directly without explicit scale decoupling) and **Our Full Model**.

1) *Experimental Design*: The core hypothesis is that while a standard regression head (TempoDepthVLM (w/o Scale)) can learn relative scene structures, it struggles to generalize metric scale to unseen domains due to the domain gap in camera intrinsics. We compare the two models under the **Absolute Mode (Unscaled)** setting on the zero-shot NYU Depth V2 dataset.

2) Quantitative Analysis:

- **Structural Consistency**: In terms of AbsRel (with alignment), both models perform comparably. This indicates that the visual encoder extracts sufficient semantic features for structural understanding in both architectures.
- **Metric Accuracy**: A significant performance gap is observed in the unscaled RMSE metric. TempoDepthVLM (w/o Scale) suffers from severe scale drift in the zero-shot setting, whereas Our Model maintains a low absolute error. This demonstrates that the dedicated Scale Head successfully disentangles global scale information from local geometric features, enabling robust zero-shot metric depth estimation.



GT Response
A wooden desk and chairs are visible...
Occluded (Baseline)
A piece of furniture, possibly a cabinet...
Injected (Ours)
A wooden desk and chairs are present, with...



GT Response
The image shows a small, cluttered room with a wooden shelf or table on the left side... To the right of the shelf, there is a cardboard bag...
Occluded (Baseline)
The image appears to be taken in a dark room... The main subject of the image is a white wall with a light switch on it...
Injected (Ours)
The image depicts a small, dimly lit room with a single piece of furniture, a desk or table... The desk appears to be made of a light-colored material...

Fig. 5. Qualitative comparisons in a single column. **Top**: Standard occlusion where our method correctly identifies the desk. **Bottom**: A low-light scenario where the baseline hallucinates a wall, but our method recovers the furniture details.

TABLE IV
ABLATION STUDY ACROSS THREE DATASETS. WE COMPARE THE BASELINE *TempoDepthVLM (w/o Scale)* WITH OUR PROPOSED *Full Model*. NOTE THAT FOR SCANNET, WE APPLY TEMPORAL INITIALIZATION TO HANDLE INTRINSIC VARIATIONS, WHILE NYU AND TUM ARE EVALUATED IN A PURE ZERO-SHOT SETTING.

Dataset	Method	Scale Aligned AbsRel ↓ RMSE ↓	Absolute AbsRel ↓ RMSE ↓
ScanNet*	TempoDepthVLM (w/o Scale)	0.143 0.370	0.316 0.652
	Ours (Full Model)	0.123 0.352	0.242 0.535
NYU Depth V2	TempoDepthVLM (w/o Scale)	0.325 1.008	0.555 2.023
	Ours (Full Model)	0.324 1.006	0.349 1.107
TUM RGB-D	TempoDepthVLM (w/o Scale)	0.294 0.916	0.611 1.639
	Ours (Full Model)	0.240 0.898	0.322 0.957

* Evaluated using temporal initialization (first 2 frames) for scale calibration.

V. DISCUSSION

A. Analysis of Results

The significant improvement in depth estimation demonstrates that VLM embeddings contain implicit 3D information that can be extracted through appropriate adapter training. The GRU memory mechanism is crucial for maintaining consistent depth predictions across frames, as it allows the model to leverage temporal context rather than making independent predictions for each frame.

The occlusion robustness results highlight the effectiveness of our Direct Feature Injection mechanism. Under severe occlusion, the baseline VLM suffers from catastrophic semantic loss or generates hallucinations (e.g., misidentifying a chair as a cabinet). Our memory injection mechanism successfully retrieves relevant historical features and restores correct semantic understanding.

B. Limitations

Despite the improvements, our method has several limitations:

- Depth accuracy still lags behind state-of-the-art dedicated depth estimation networks (e.g., DepthAnything [6]). Our method produces coarse 3-region depth rather than dense per-pixel predictions, and is designed for relative depth understanding rather than precise metric reconstruction.
- Experiments are conducted on a single backbone (Qwen2-VL-2B). Generalization to other larger VLM backbones needs further investigation.
- The memory injection mechanism requires a “warm-up” period to accumulate reliable memories, which may limit effectiveness at the beginning of a video sequence or after scene transitions.

VI. CONCLUSION

We presented **TempoDepth-VLM**, a parameter-efficient adapter designed to empower frozen Vision-Language Models with temporal consistency and metric depth awareness. By integrating a Transformer encoder with GRU-based long-term memory and a novel Shape-Scale Decoupled Head, our adapter enables VLMs to maintain stable geometric understanding across video frames.

Our experiments demonstrate that our method drastically outperforms the baseline on the ScanNet test set, achieving remarkable gains in both structural consistency and metric accuracy. Furthermore, the model exhibits strong zero-shot generalization on unseen datasets (NYU Depth V2 and TUM RGB-D) even without test-time alignment, validating the effectiveness of our scale decoupling strategy in learning generic scale priors. Qualitative results further confirm the robustness of our Direct Feature Injection mechanism in handling occlusions and maintaining temporal stability.

These findings suggest that lightweight adapters can effectively extend static-image VLMs to become more 3D-aware and temporally consistent. Future directions include: (1) incorporating ego-motion and pose estimation for comprehensive 3D scene understanding, (2) evaluating the adapter

on larger VLM backbones for plug-and-play applicability, and (3) deploying the framework in downstream navigation and assistive robotics tasks.

VII. TEAM CONTRIBUTIONS

Yi-Zhe Lu: Architecture design; temporal adapter + GRU memory implementation; training integration.

Yu-Hsiang Chuang: Temporal adapter + GRU memory implementation; training integration.

You-Cen Tsai: Occlusion robustness analysis and demos.

Pin-Yu Wu: Survey related works ; prepare slides.

All members: Discussed ideas, reviewed code, refined the report, and prepared the final presentation.

REFERENCES

- [1] B. Chen, Z. Xu, S. Kirmani, et al., “SpatialVLM: Endowing vision-language models with spatial reasoning capabilities,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024.
- [2] C. Yang, et al., “MMA: Multi-modal adapter for vision-language models,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024.
- [3] J. Zhang and G. Lu, et al., “Vision-language embodiment for monocular depth estimation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025.
- [4] K. Cho, B. van Merriënboer, C. Gulcehre, et al., “Learning phrase representations using RNN encoder-decoder for statistical machine translation,” in *Proc. Conf. Empirical Methods Natural Language Process. (EMNLP)*, 2014.
- [5] A. Dai, A. X. Chang, M. Savva, et al., “ScanNet: Richly-annotated 3D reconstructions of indoor scenes,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017.
- [6] L. Yang, B. Kang, Z. Huang, et al., “Depth anything: Unleashing the power of large-scale unlabeled data,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024.